Lab 6: Code and Branch Coverage

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1. Code (line) coverage

Tests: CodeCoverageTest_cloud_storage_tier.py. Line coverage of cloud_storage_tier.py = **100%** (57/57 statements).

```
$ coverage run -m pytest CodeCoverageTest_cloud_storage_tier.py
============================= test session starts ==============================
platform linux -- Python 3.12.3, pytest-9.2.0, pluggy-1.7.0
rootdir: /tmp/L6_L7_12_cloud_storage_tier-main/Lab 6
plugins: platformdirs-4.13.0, hypothesis-6.170.15, cov-7.1.0
collected 9 items

CodeCoverageTest_cloud_storage_tier.py .........                         [100%]

============================== 9 passed in 0.22s ===============================

$ coverage report -m
Name                                    Stmts   Miss  Cover   Missing
---------------------------------------------------------------------
CodeCoverageTest_cloud_storage_tier.py      33      0   100%
cloud_storage_tier.py                       57      0   100%
---------------------------------------------------------------------
TOTAL                                       90      0   100%
```

2. Branch coverage

Tests: BranchCoverageTest_cloud_storage_tier.py. Line coverage = **100%**, branch coverage = **100%** (40/40 branches, 0 partial).

```
$ coverage run --branch -m pytest BranchCoverageTest_cloud_storage_tier.py
============================= test session starts ==============================
platform linux -- Python 3.12.3, pytest-9.2.0, pluggy-1.7.0
rootdir: /tmp/L6_L7_12_cloud_storage_tier-main/Lab 6
plugins: platformdirs-4.13.0, hypothesis-6.170.15, cov-7.1.0
collected 33 items

BranchCoverageTest_cloud_storage_tier.py ............................. [ 93%]
..                                                                    [100%]

============================== 33 passed in 0.23s ==============================

$ coverage report -m
Name                                     Stmts   Miss Branch BrPart  Cover   Missing
------------------------------------------------------------------------------------
BranchCoverageTest_cloud_storage_tier.py     105      0      0      0   100%
cloud_storage_tier.py                         57      0     40      0   100%
------------------------------------------------------------------------------------
TOTAL                                        162      0     40      0   100%
```